Recent fat intake modulates fat taste sensitivity in lean and overweight subjects.

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OBJECTIVE: To evaluate the effects of a high-fat and low-fat diet on taste sensitivity to oleic acid (C18:1) in lean and overweight/obese (OW/OB) subjects. DESIGN: Randomized cross-over dietary intervention involving the consumption of a high-fat (>45% fat) and low-fat (<20% fat) diet, both consumed over a 4-week period. SUBJECTS: A total of 19 lean, mean age 33±13 years, mean body mass index (BMI) 23.2±2.2 kg m(-2) and 12 OW/OB, mean age 39.5±3 years, mean BMI 28±2.6 kg m(-2), subjects participated in the study, which measured taste thresholds for C18:1, fat perception and hedonic ratings for regular (RF) and lowered-fat (LF) foods before, and following consumption of a high- and low-fat diet. RESULTS: Consumption of the low-fat diet increased taste sensitivity to C18:1 among lean and OW/OB subjects ($P<0.05$) and increased the subjects ability to perceive small differences in the fat content of custard ($P=0.05$). Consumption of the high-fat diet significantly decreased taste sensitivity to C18:1 among lean subjects ($P<0.05$), with no change in sensitivity among OW/OB persons ($P=0.609$). The hedonic ratings for several RF and LF foods differed following the diets. CONCLUSION: Alterations in the fat content of the diet modulated taste sensitivity to C18:1 among lean subjects, which was increased following a 4-week period of fat restriction and attenuated following the high-fat diet. The failure of the high-fat diet to alter fatty acid taste thresholds among OW/OB subjects suggests that these individuals were 'adapted' to high-fat exposure, perhaps because of differences in habitual fat consumption. Taken together, these data suggest that excessive dietary fat attenuates nutrient sensing epithelia response in the oral cavity, which could be associated with changes in diet and weight status.